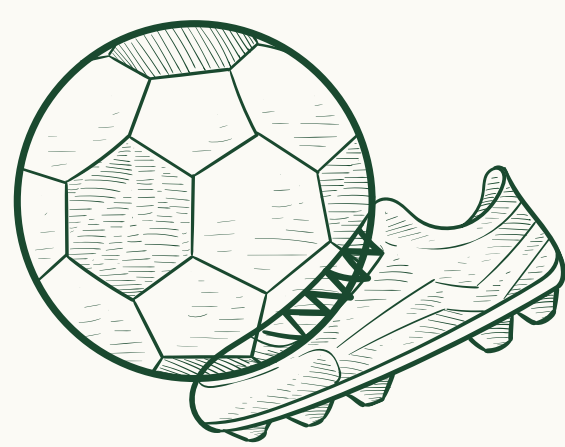


FULBO

FIELD UNIFIED LOCALIZATION AND BALL/PLAYER OBSERVATION



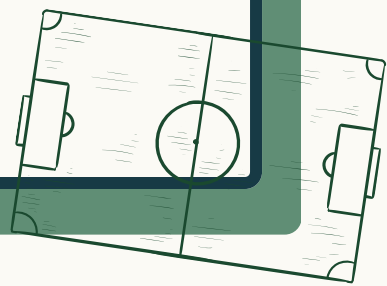
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I308 – Computer Vision
Universidad de San Andrés – Spring 2025

INTRODUCTION

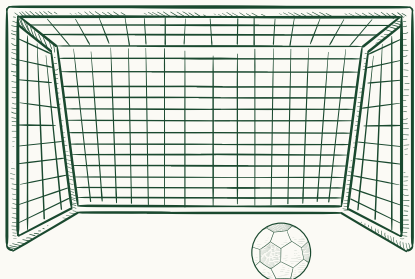
Soccer match analysis requires detecting and tracking players and the ball in scenes characterized by fast motion, frequent occlusions, and variations in team uniforms.

This project presents a system based on YOLOv8, ByteTrack, DINOv2, UMAP, K-Means, and homography estimation to interpret video sequences and reconstruct player positions on the soccer field.

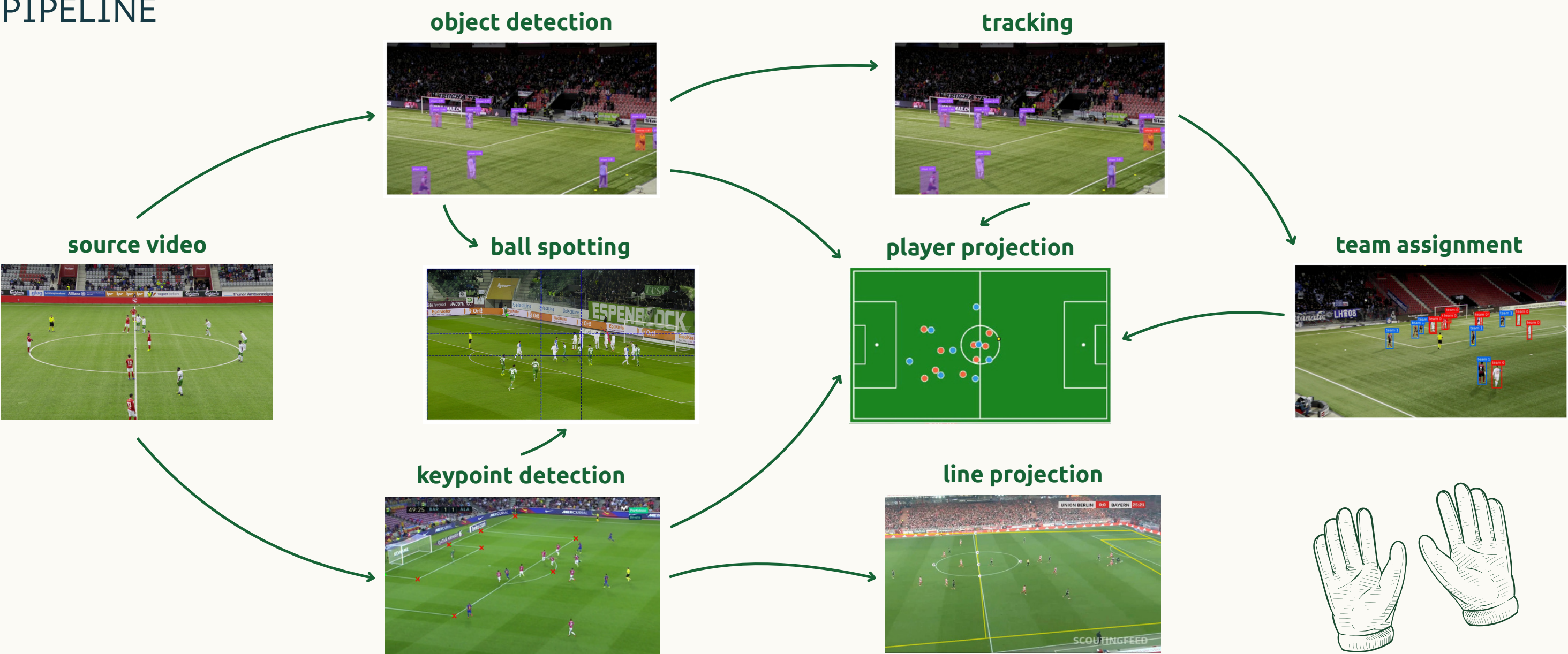


GOALS

- Detect players, the referee, and the ball in each video frame.
- Maintain persistent tracking of each player and the ball throughout the sequence.
- Differentiate the two teams using visual embeddings and clustering.
- Detect keypoints to estimate a homography and project player positions onto a field minimap.



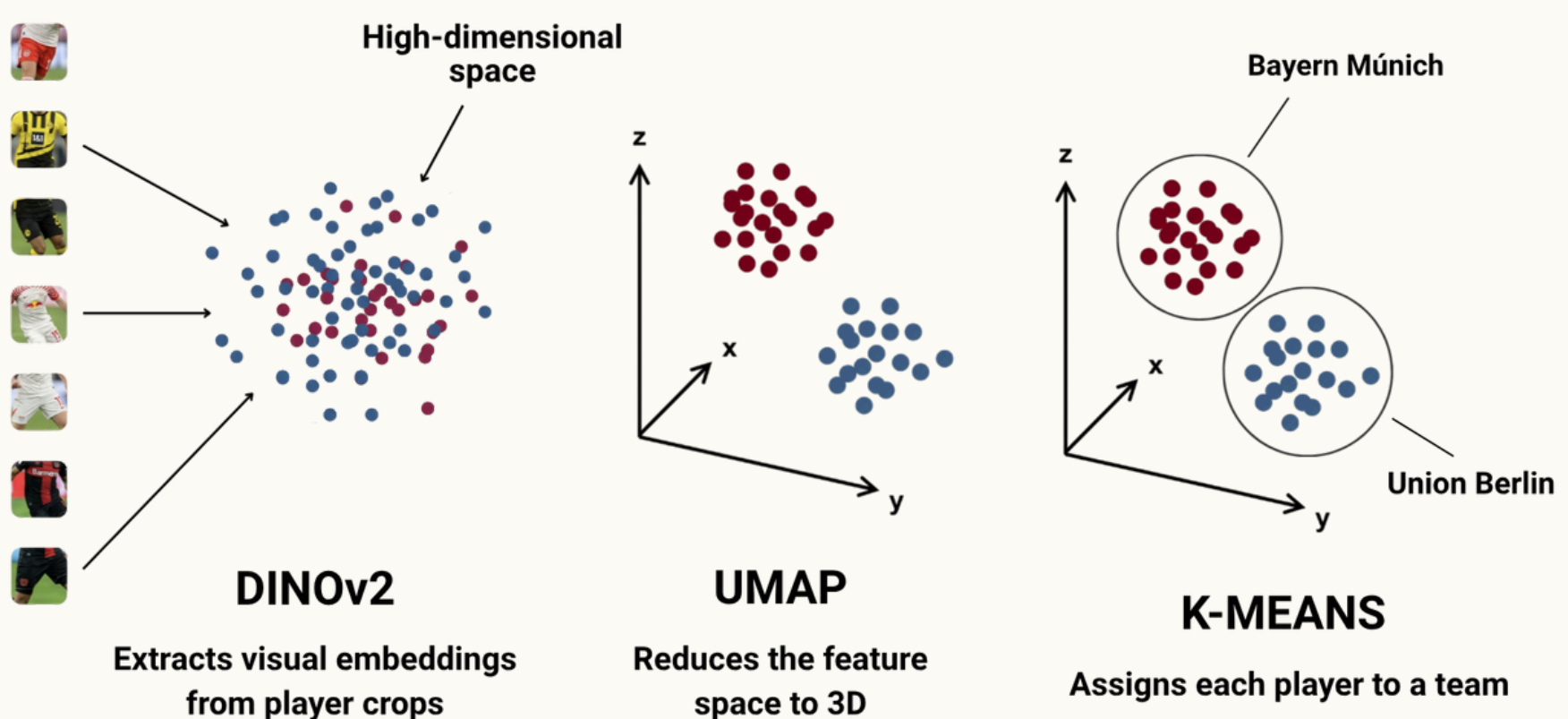
PIPELINE



- Object detection (YOLOv11).
- ByteTrack.
- Visual feature extraction (DINOv2).
- Dimensionality reduction (UMAP).
- Team clustering (K-Means).
- Keypoint detection (YOLOv8-Pose).
- Homography.
- Tactical minimap.

KEY RESULTS

- **GLOBAL (player + referee):**
Precision = 0.894
Recall = 0.880
- **PLAYER:**
Precision = 0.891
Recall = 0.908
- **REFEREE:**
Precision = 0.942
Recall = 0.586
- **KEYPOINTS (29):**
Precision = 0.805
Recall = 0.856



FUTURE WORK

- Incorporate player segmentation.
- Explore more advanced tracking methods by combining appearance and motion information.
- Develop a dedicated goalkeeper detection module.
- Develop an action detection model.
- Integrate tactical analysis modules.
- Build larger and more diverse datasets, including different leagues, jersey designs, and visual conditions.
- Evaluate homography estimation based on PnP, with the potential to extend the system to 3D through ball triangulation.



References: SoccerNet. (2022). SoccerNet datasets — tracking & calibration. / Roboflow. (2024, junio 10). Camera Calibration for Sports Computer Vision [Blog post]. Roboflow Blog. / Adit-jain. (2023). Soccer_Analysis: SoccerNet_Keypoints module [Software/data]. / NikolasEnt. (2023). soccernet-calibration-sportlight [Software].